
PDS data acquisition and local trigger conditions for VD LE

single photoelectron activity expectations
and mitigation strategies

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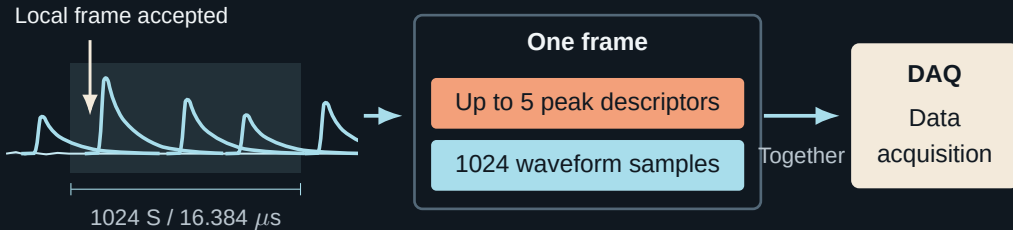
17 September 2026

DUNE
DEEP UNDERGROUND
NEUTRINO EXPERIMENT

 **Fermilab**

Current DAPHNE gateway: 1024 samples per frame

DAPHNE samples each channel continuously at the AFE level:
 $62.5 \text{ MSPS} \times 14 \text{ bits} = 875 \text{ Mbit/s/channel}$.



Frame = saved samples + peak summaries. A descriptor reports pulse timing and size.

Baseline: `daphne-os` + `daphne-firmware`; self-trigger 3f17f1b.

Five summaries may not describe every peak

Waveform sent

One 1024-sample frame



Peak descriptors sent

Peak 1

Peak 2

Peak 3

Peak 4

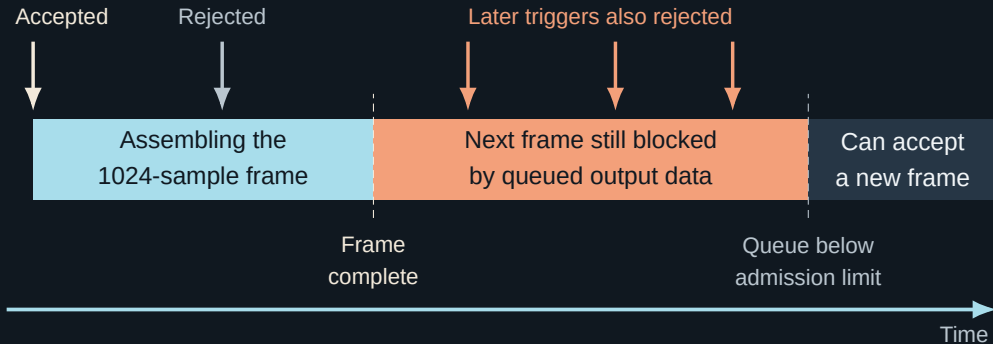
Peak 5

Peaks 6 and 7:
no descriptor

All seven peaks remain in the waveform.
Only five have a summary.

Illustrative resolved peaks; numbers identify peaks, not photoelectron counts.

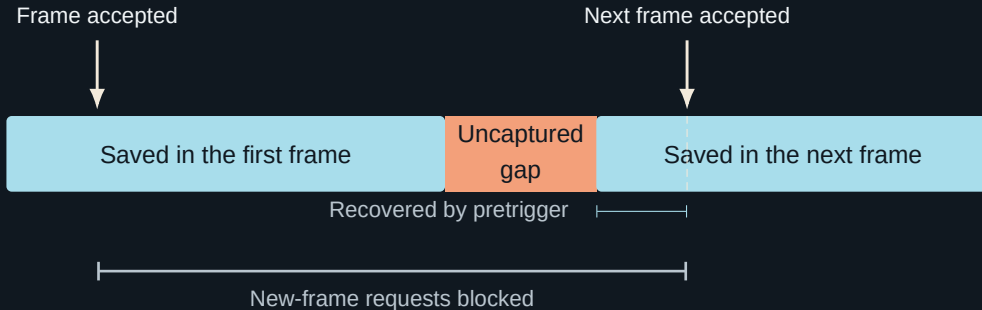
Dead time can continue after the frame is complete



A full output FIFO can block the next frame long after assembly ends.

Illustrative congested case, not to scale. The extra wait depends on how quickly shared readout drains the queue.

Count the time when signal cannot be saved



Dead time = blocked, uncaptured time / observation time.

A rejected request is not necessarily lost light: its signal may already be in a saved frame.

We needed an activity estimate for each VD channel

Without an agreed vertical-drift (VD) forecast,
we simulated atmospheric argon and gamma backgrounds.

87 kHz

per channel at 0.7 PE

VD cathode estimate, before firmware losses.

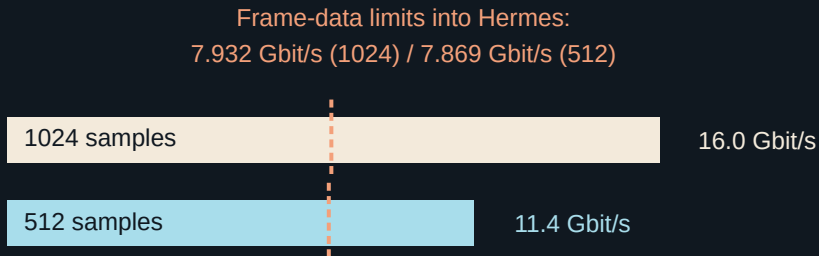
A candidate is a pulse
above the chosen threshold.

Study reference: intrinsic backgrounds; eight LArSoft windows; threshold > 0.7 photoelectrons.

87 kHz/channel is the rounded cathode mean, not a uniform rate across VD. Gammas were studied separately.

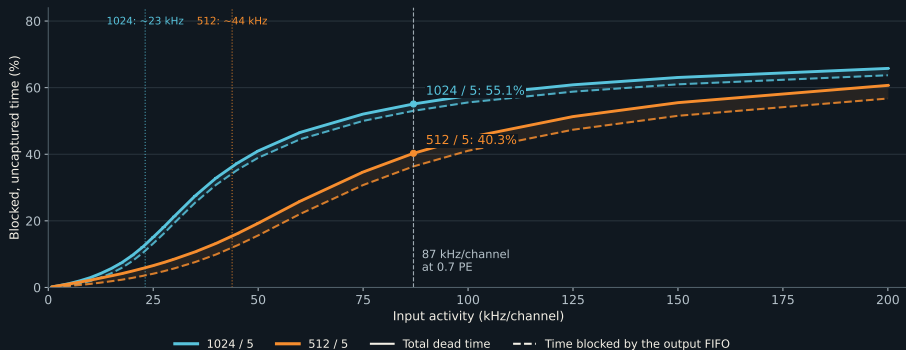
One link is still the bottleneck

Current agreement: one link per board.



32 cathode channels at about 87 kHz/channel, threshold > 0.7 PE. Demand after the local busy gate (33.62 / 46.53 kframes/s/channel), before shared FIFO losses. Red: frame-data limits with output ready.

Limited output capacity drives dead time up



Dotted markers: mean offered load reaches the ~7.9 Gbit/s shared-readout budget. Bursts cause loss earlier.

32 channels/board; one link; matched Poisson inputs. Reference: 87 kHz/channel at 0.7 PE.

Finite shared readout into Hermes; dashed curves count FIFO-blocked, uncaptured time. Additional Hermes/UDP stalls are not simulated.

More links buy capacity, with an infrastructure cost

1 link

7.869 Gbit/s

Current agreement

2 links

15.738 Gbit/s

Option to study

4 links

31.475 Gbit/s

Option to study

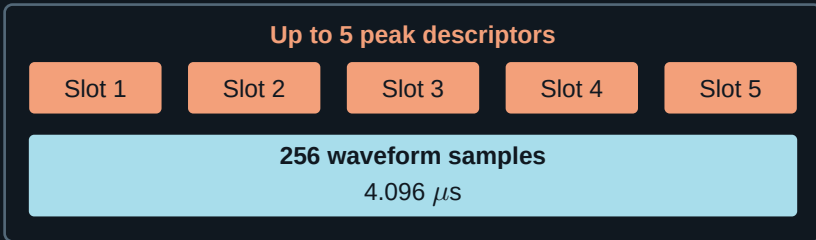
Waveforms and peak descriptors share this bandwidth.

The price: more DAQ servers and power capacity
in the Central Utility Cavern (CUC).

512-sample sender budget into Hermes, including scan/pause cycles.
Balanced load, output ready; network overhead excluded.

Four links meet the 256 / 5 frame-data budget

One accepted frame

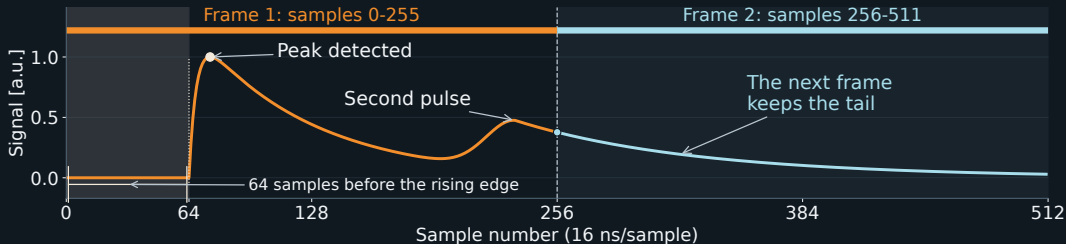


Each link needs 7.875 Gbit/s. Gapless readout allows 8 Gbit/s.

Frame data into Hermes: 32 channels across four balanced links.
Full-load budget; sustained Ethernet/DAQ delivery still needs validation.

256 / 5 loss under the VD activity trace has not been measured.

The next frame keeps the pulse's tail

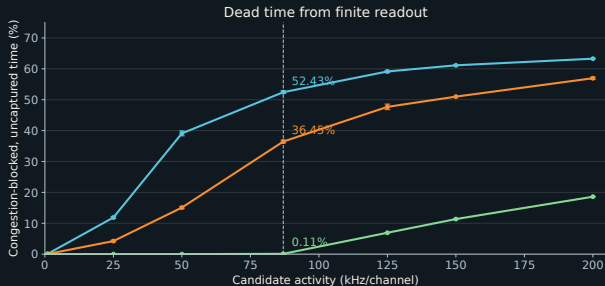


Still active? Continue from sample 256.
No new trigger is needed.

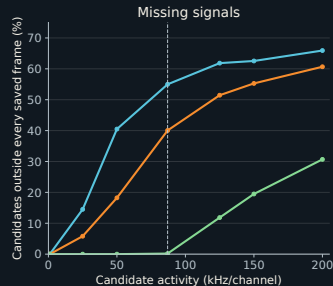
Illustration adapted from the grouped32 study. Continuation enabled, with buffer space available.
Consecutive saved samples; queued assembly and readout.

Two links look promising at the expected activity

32 channels: matched candidate timestamps — pilot comparison



— 1024 samples / 5 descriptors / 1 link — 512 samples / 5 descriptors / 1 link — 256 samples / 5 descriptors / 2 links



87 kHz/channel: reference at 0.7 PE

256: grouped RTL + portable XPM; 1024/512: C++ models. Finite readout into Hermes; full UDP/MAC stalls not modeled.
Two seeds; 0.32 ms warmup + 0.96 ms measured per seed. 256 continuation enabled; synthetic 32-tick pulses. Bars: SEM only.

At 87 kHz/channel: 256 / 5 / 2 gives 0.11% congestion dead time in this pilot.